

Please make sure that you show all of the steps, not just the answer.

1) Use the definition of the derivative: $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find the derivative of $f(x) = x^2 + 5x$ when $x = 4$.

$$f(4) = (4)^2 + 5(4)$$

$$= 16 + 20$$

$$= 36$$

Point: (4, 36)

$$\frac{(x^2 + 5x) - (36)}{(x) - (4)}$$

factor numerator
 $\hookrightarrow (x+9)(x-4)$

$$\frac{(x+9)(\cancel{x-4})}{\cancel{x-4}}$$

$$\hookrightarrow (x+9)$$

$$\lim_{x \rightarrow 4} (4) + 9$$

$$= 13$$

$$f'(4) = 13$$

2) Find the equation of the line tangent to $f(x) = \sqrt{x}$ when $x = 4$.

Rewrite $f(x) = x^{1/2}$

Power rule $\rightarrow = \frac{1}{2}x^{-1/2}$

$$= \frac{1}{2\sqrt{x}}$$

$$F(4) = \sqrt{4}$$

$$= 2$$

Point (4, 2)

$$f'(4) = \frac{1}{2\sqrt{4}}$$

$$= \frac{1}{2(2)}$$

Slope $m = \frac{1}{4}$

$$y - (2) = \left(\frac{1}{4}\right)(x - (4))$$

$$y = \frac{1}{4}x - 1 + 2$$

$$y = \frac{1}{4}x + 1$$

3) Find the points (x, y) where $y'(x) = 0$ for the function $y(x) = x^3 - 5x^2 + x + 6$.Differentiate $\rightarrow$ Set = 0

$$3x^2 - 10x + 1 = 0$$

Solve for x

$$x = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(3)(1)}}{2(3)}$$

$$= \frac{10 \pm \sqrt{100 - 12}}{6}$$

$\left. \begin{array}{l} \sqrt{4 \cdot 22} \\ \downarrow \\ 2\sqrt{22} \end{array} \right\}$

$$= \frac{10 \pm 2\sqrt{22}}{6}$$

$$= \frac{2(5 \pm \sqrt{22})}{\cancel{6} 3} \div \frac{2}{2}$$

$$x = \frac{5 \pm \sqrt{22}}{3}$$

Points

$$\left[\left(\frac{5 + \sqrt{22}}{3} \right), y \left(\frac{5 + \sqrt{22}}{3} \right) \right]$$

$$\left[\left(\frac{5 - \sqrt{22}}{3} \right), y \left(\frac{5 - \sqrt{22}}{3} \right) \right]$$

4) For any given function, explain the difference between $y'(x) = 0$ and $y'(0)$.

$y'(x) = 0$ is an equation.

"Find all values of x where the derivative equals zero"
 ↳ where is slope zero?

$y'(0)$ is a value.

"Take the derivative, then plug in $x=0$ "
 ↳ this gives the slope of the function at $x=0$

5) Find all values of x so that $f'(x) = 0$ for the function $f(x) = (x+5)^5(x-1)^6$

$$A = (x+5)^5 \quad A' = 5(x+5)^4$$

$$B = (x-1)^6 \quad B' = 6(x-1)^5$$

Product Rule
 $f(x) = A'B + AB'$

$$f'(x) = \overset{A'}{5(x+5)^4} \cdot \overset{B}{(x-1)^6} + \overset{A}{(x+5)^5} \cdot \overset{B'}{6(x-1)^5}$$

$$= (x+5)^4 (x-1)^5 \left[\underbrace{5(x-1)}_{5x-5} + \underbrace{6(x+5)}_{6x+30} \right]$$

$$[11x + 25]$$

$$f(x) = (x+5)^4 (x-1)^5 (11x + 25) = 0$$

then set equal to zero ↗

$$1) (x+5)^4 = 0$$

$$x = -5$$

$$2) (x-1)^5 = 0$$

$$x = 1$$

$$3) 11x + 25 = 0$$

$$11x = -25$$

$$x = -\frac{25}{11}$$

$$x = -5, 1, -\frac{25}{11}$$

6) Find the equation of the line tangent to $f(x) = x \sin(x)$ when $x = \pi$.

$$f'(x) = \overset{A'}{1} \overset{B}{\sin x} + \overset{A}{x} \overset{B'}{\cos x}$$

$$= \sin x + x \cos x$$

Product Rule

$$f(\pi) = \pi \sin(\pi)$$

$$= \pi(0)$$

$$= 0$$

Point: $(\pi, 0)$

Find the Slope at $x = \pi$

$$f'(\pi) = \underbrace{\sin(\pi)}_0 + \pi \underbrace{\cos(\pi)}_{-1}$$

$$= 0 + \pi(-1)$$

$$= -\pi$$

$$M = -\pi$$

use
Point Slope form

$$y - 0 = -\pi(x - \pi)$$

$$y = -\pi(x - \pi) + 0$$

$$y = -\pi x + \pi^2$$